MILITARY SPECIFICATION

CASTABLE MIX, REFRACTORY, THERMAL INSULATING

1. SCOPE

1.1 This specification covers thermal insulating refractory castable mix for use as a back-up insulation for refractory furnace linings of boilers.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL
UU-S-48 - Sacks, Paper, Shipping.
PPP-B-35 - Bags: Textile, Shipping, Burlap, Cotton and Waterproof Laminated.

STANDARDS

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, N.Y.)

3. REQUIREMENTS

3.1 Material.- The castable refractory shall be composed of a mixture of heat-resistant materials and a bond material. It shall mix readily with water and shall be of such consistency when mixed with water that it can be cast into a sound wall reasonably free of voids.

3.2 Time for air set.- The length of time for air setting shall not be greater than 24 hours. At the end of this period, the mix shall support its own weight and maintain its originally cast shape (see 4.4.1).

3.3 Density.- The density of the castable refractory shall not exceed 45 pounds per cubic foot after being heat soaked at 2,000°F. for 5 hours (see 4.4.2).

3.4 Permanent linear change.- The permanent linear change shall not exceed 3.0 percent shrinkage after being heat soaked at 2,000°F. for 5 hours (see 4.4.3).

3.5 Refractoriness.- The castable refractory shall be in satisfactory condition without evidence of fusion or softening upon completion of the test specified in 4.4.4 and shall retain its cast shape without crumbling when removed from the panel assembly.

FSC 9350

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3.6 Thermal conductance.- The thermal conductance of the mix shall be such that the temperature drop through and block cast from the mix shall be not less than 1,800°F. when the hot face of the block is maintained at 2,500°F. (see 4.4.4).

3.7 Workmanship.- Workmanship shall be first class in every respect.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory satisfactory to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling for quality conformance inspection.- Sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated. For purposes of sampling, an inspection lot for quality conformance inspection shall consist of all material offered for delivery at one time.

4.3 Inspection of the castable mix.-

4.3.1 Examination of the castable mix.- Examination of the castable mix shall be made in accordance with the classification of defects, inspection levels and acceptable quality levels (AQLs) set forth below. The lot size, for purpose of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of filled containers of castable refractory mix for examination in 4.3.1.1, 4.3.1.2 and 4.3.1.3.

4.3.1.1 Examination of castable mix container for defects in appearance, workmanship, closures and markings.- The sample unit for this examination shall be one filled container.

Examine	Defect
Appearance and workmanship	Not type or size specified. Evidence of sifting or leaking of contents. Bag laminations not as specified. Scuff marks, torn or punctured.
Closures	End closure not as specified.
Markings	Omitted, illegible, incorrect, incomplete or not in accordance with contract requirements.

4.3.1.2 Examination of castable mix container for defects in appearance of refractory castable mix.- The sample unit for this examination shall be one filled container.

Examine	Defect
Appearance	Material not as specified. Presence of foreign material. Presence of large particles, lumps or caked material.

4.3.1.3 Examination of castable mix for defects in net contents. The sample unit for this examination shall be one filled container. The average net contents shall be not less than the specified or indicated quantity. No less than five sample units shall be drawn for this examination.

4.3.1.4 Inspection levels and acceptable quality levels (AQLs) for examinations.- The inspection levels for determining the sample size and acceptable quality levels (AQLs) expressed in defects per 100 units shall be as follows:

Examination Paragraph	Inspection Level	AQL
4.3.1.1	I	2.5
4.3.1.2	L-6	2.5
4.3.1.3	L-4	- -

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4.3.2 Testing of the castable mix.- The castable mix shall be tested for the applicable characteristics as indicated in table I for each lot presented for examination. The sample unit shall be 15 pounds of the refractory castable mix. The inspection level for determining sample size shall be L-2 except that no less than five sample units shall be randomly selected throughout the lot. No more than one sample shall be drawn from any one container. The test sample shall consist of a 75-pound composite of the sample units. Two or more determinations, as applicable, shall be performed on the composite. There shall be no evidence of failure to meet the specified requirements.

Table I - Instructions for testing.

Characteristics	Specification Reference		Number Determinations per (composite)	Results Reported As	
	Requirement	Test Method		Pass or Fail 1/	Numerically To Nearest 2/
Airsetting	3.2	4.4.1	2	X	-
Density	3.3	4.4.2	Avg of 2	-	0.1 lb. per cu. ft.
Permanent linear change	3.4	4.4.3	Avg of 2	-	0.1 percent
Refractoriness	3.5	4.4.4	2		
Fusion				X	-
Softening				X	-
Crumbling				X	-
Thermal conductance	3.6	4.4.4	Avg of 2	-	10°F.

1/ If failure is indicated, report either description of failure or numerical point of failure.
2/ Test reports shall include all values on which average results are based.

4.4 Tests.-

4.4.1 Time for air set.- The test material shall be thoroughly mixed with water to the consistency of stiff concrete. This mixture shall then be poured into 9 by 4-1/2 by 2-1/2-inch steel molds and shall be placed at once in a chamber having constant atmosphere of 90°F. and 90 percent humidity. At the end of 24 hours the mold shall be removed.

4.4.2 Density.- The test bricks prepared as specified in 4.4.1 shall be returned to the constant humidity chamber for an additional 24 hours of curing. The test bricks shall then be removed and oven dried for 12 hours at temperature of 220°F. Then the test bricks shall be measured. The test bricks shall be subjected to a 5 hour heat soak at 2,000°F. and then allowed to cool in the closed furnace to room temperature. The test bricks shall then be weighed. The density shall be computed from the oven dry volume and the weight after firing at 2,000°F.

4.4.3 Permanent linear change.- The permanent linear change shall be computed from the molded length of the test bricks prepared as specified in 4.4.1 and the fired length of these bricks after being tested as specified in 4.4.2.

4.4.4 Refractoriness and thermal conductance.- The test material shall be thoroughly mixed with water to the consistency of stiff concrete and cast into a block 18 by 18 by 3-1/2 inches. The block shall be allowed to remain in the mold for 24 hours at atmospheric temperatures and then removed from the mold and allowed to air dry for an additional 24 hours. The test block shall be built into a panel. The panel shall be composed of the 3-1/2-inch thick block faced with 2-1/2 inches of firebrick and backed with a 1/8 inch thick steel plate. The firebrick face of the panel shall be exposed to the furnace atmosphere. The interface of the block and the firebrick shall be maintained at a temperature of 2,500°F. for 24 hours. Then the temperature drop through the cast block shall be determined. The panel shall be allowed to cool to room temperature. Then the test block shall be examined for evidence of fusion, softening or crumbling.

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5. PREPARATION FOR DELIVERY

5.1 Packing.-

5.1.1 Level A.- Castable refractory shall be packed in 50 pound quantities in multi-wall shipping sacks construction number 5X or textile laminated shipping bags conforming to UU-S-48 and PPP-B-35, respectively. The specific container selected shall incorporate a liner or ply to provide required moisture protection.

5.1.2 Level B.- Castable refractory shall be packed in 50 pound quantities in multi-wall shipping sacks construction number 1W or textile laminated shipping bags conforming to UU-S-48 and PPP-B-35, respectively. The specific container selected shall incorporate a liner or ply to provide required moisture protection.

5.1.3 Level C.- Castable refractory shall be packed in a manner to insure carrier acceptance and safe delivery to destination at the lowest applicable rate. Containers shall comply with the Uniform Freight Classification Rules or other carrier regulations applicable to the mode of transportation.

5.2 Marking.- In addition to any special marking required in the contract or order, marking for shipment shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Applicable level of packing required (see 5.1).

Notice.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Preparing activity:
Navy - Bureau of Ships
(Project 9350-N013Sh)

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SPECIFICATION ANALYSIS SHEET

NAVSHIPS-4863 (8-61)

BUDGET BU. NO. 45-R309

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Bureau of Ships

This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this form will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to Bureau of Ships, Specifications and Standardization Branch, Washington 25, D.C.

SPECIFICATION

ORGANIZATION	CITY	STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   a. GIVE PARAGRAPH NUMBER AND WORDING

   b. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID.

3. IS THE SPECIFICATION RESTRICTIVE? ☐ YES ☐ NO IF THE ANSWER IS "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification.)* PLACE THIS FORM AND PAPERS IN AN ENVELOPE AND SEND TO THE BUREAU.

SUBMITTED BY *(Print name and activity)*	DATE

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